

Math 196 Lecture Notes: Week 3

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0.1 Review of Matrix Multiplication

Last week, we defined the matrix product AB when A is an $r \times q$ matrix, and B is a $q \times p$ matrix. This is a $r \times p$ matrix, which represents the composite map

$$\mathbb{R}^p \xrightarrow{B} \mathbb{R}^q \xrightarrow{A} \mathbb{R}^r$$

taking $v \in \mathbb{R}^p$ to $A(Bv)$. If we write B in terms of columns,

$$B = \begin{bmatrix} \vdots & & \vdots \\ v_1 & \dots & v_p \\ \vdots & & \vdots \end{bmatrix}$$

Then we can compute AB by applying A to each column:

$$AB = \begin{bmatrix} \vdots & & \vdots \\ Av_1 & \dots & Av_p \\ \vdots & & \vdots \end{bmatrix}$$

0.2 Plan for this week

We will first introduce the dot product, another operation on vectors. This will play a major role later in the course but for now it serves two purposes for us. First, it will give us another way of describing the computation of a matrix product AB , and second, it will give us a way to describe reflections across arbitrary lines in the plane.

After this, we will introduce inverses of matrices and how to compute them. Our third topic this week will be an introduction to linear subspaces, linear dependence, and bases.

1 Dot Products and more Geometry

1.1 Dot Products and the Row View of Matrix Vector Multiplication

In this section we will introduce another operation on vectors, use it to give another view of matrix multiplication, and consider its geometric significance.

Given two vectors

$$v = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}, w = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$$

we define their dot product by

$$v \cdot w = x_1 y_1 + \dots + x_n y_n$$

Example 1.

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} -1 \\ 3 \end{bmatrix} = 1 \cdot (-1) + 2 \cdot 3 = 5$$

We have seen this operation in matrix vector multiplication: In the simplest case, consider a $1 \times n$ matrix

$$A = [x_1 \quad \dots \quad x_n]$$

This takes as input vectors in $\mathbb{R}^n$ and outputs vectors in $\mathbb{R}^1$, which we can view just as scalars. Specifically

$$\begin{bmatrix} x_1 & \dots & x_n \end{bmatrix} \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} = x_1 y_1 + \dots + x_n y_n$$

is the dot product of the vector corresponding to A with the input vector. For any arbitrary matrix A , each entry in Av is the dot product of the corresponding row of A with v :

$$\begin{bmatrix} \dots & r_1 & \dots \\ & \vdots & \\ \dots & r_m & \dots \end{bmatrix} v = \begin{bmatrix} r_1 \cdot v \\ \vdots \\ r_m \cdot v \end{bmatrix}$$

where we view the rows r_i as vectors by turning them.

Example 2. We can compute the product

$$\begin{bmatrix} 2 & 0 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

by computing dot products:

$$\begin{bmatrix} 2 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 2, \quad \begin{bmatrix} -1 \\ 3 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 2$$

so our answer is

$$\begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

The dot product satisfies some algebraic properties, like the other products we have seen:

$$(u + v) \cdot w = u \cdot w + v \cdot w$$

$$u \cdot (v + w) = u \cdot v + u \cdot w$$

$$(av) \cdot w = a(v \cdot w) = v \cdot (aw)$$

This product is commutative:

$$v \cdot w = w \cdot v$$

1.1.1 Entries of a matrix product as dot products

This interpretation of matrix-vector multiplication gives us a concise way of representing the elements of a product of two matrices: Consider

$$A = \begin{bmatrix} \dots & r_1 & \dots \\ & \vdots & \\ \dots & r_m & \dots \end{bmatrix}, \quad B = \begin{bmatrix} \vdots & & \vdots \\ v_1 & \dots & v_n \\ \vdots & & \vdots \end{bmatrix}$$

The the entry of AB in column i , row j is the j th entry in Av_i , which is exactly the dot product $r_j \cdot v_i$, again viewing the r_j s as vectors. [Draw picture, highlighting rows and columns] [Maybe do an example here.]

1.2 Geometry and the Dot Product

I will now discuss a bit of the geometry of the dot product, in order to describe more linear transformations of the plane that can be expressed in terms of it. For now this will be a bit terse, but we will return to these points in more detail later in the course, when we discuss orthogonal matrices. For now, all that we will need is that two vectors in $\mathbb{R}^2$ are at a right angle to each other exactly when their dot product is 0.

1.2.1 Projection onto a line

Suppose we have vectors $v, w \in \mathbb{R}^2$, with $w \neq 0$. If we let a vary, the vectors aw sweep out a line. We would like to compute the projection of v onto this line, which we denote $\text{proj}_w(v)$. [Draw picture with right triangle]. This is a vector aw for some a , such that $v - aw$ is at a right angle to w . The only thing left to determine is the constant a . Using the algebraic properties of the dot product, we compute

$$0 = w \cdot (v - aw) = w \cdot v - a(w \cdot w)$$

Solving for a , we get

$$a = \frac{w \cdot v}{w \cdot w}$$

and so

$$\text{proj}_w(v) = \left(\frac{w \cdot v}{w \cdot w} \right) w$$

Using this formula, we can check that for fixed w , the function proj_w is linear:

$$\text{proj}_w(u + v) = \left(\frac{w \cdot (u + v)}{w \cdot w} \right) w = \left(\frac{w \cdot u}{w \cdot w} + \frac{w \cdot v}{w \cdot w} \right) w = \left(\frac{w \cdot u}{w \cdot w} \right) w + \left(\frac{w \cdot v}{w \cdot w} \right) w = \text{proj}_w(u) + \text{proj}_w(v)$$

$$\text{proj}_w(av) = \left(\frac{w \cdot av}{w \cdot w} \right) w = a \left(\frac{w \cdot v}{w \cdot w} \right) w = a \cdot \text{proj}_w(v)$$

[Note: in these formulas, vector and scalar quantities should be distinguished using the arrow.]

We should be able to represent this transformation with a matrix. To do this, we let

$$w = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$$

and we compute

$$\begin{aligned} \text{proj}_w(e_1) &= \frac{w_1}{w \cdot w} w = \frac{1}{w_1^2 + w_2^2} \begin{bmatrix} w_1^2 \\ w_1 w_2 \end{bmatrix} \\ \text{proj}_w(e_2) &= \frac{w_2}{w \cdot w} w = \frac{1}{w_1^2 + w_2^2} \begin{bmatrix} w_1 w_2 \\ w_2^2 \end{bmatrix} \end{aligned}$$

The projection matrix comes out to

$$\frac{1}{w_1^2 + w_2^2} \begin{bmatrix} w_1^2 & w_1 w_2 \\ w_1 w_2 & w_2^2 \end{bmatrix}$$

Example 3. Projection onto the line $y = x$, through the vector

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

is given by

$$\begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix}$$

1.2.2 Reflections

Using the formula for projection, we can also compute the reflection of a vector across the line, which I will denote $T_w(v)$

[Draw Picture]

The formula we get from the diagram is

$$T_w(v) = \text{proj}_w(v) - (v - \text{proj}_w(v)) = 2 \cdot \text{proj}_w(v) - v$$

The matrix representing this comes out to

$$\frac{2}{w_1^2 + w_2^2} \begin{bmatrix} w_1^2 & w_1 w_2 \\ w_1 w_2 & w_2^2 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Example 4. Consider the vector

$$w = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

The projection matrix is

$$\frac{1}{5} \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} 1/5 & 2/5 \\ 2/5 & 4/5 \end{bmatrix}$$

The reflection matrix is

$$\begin{bmatrix} 2/5 & 4/5 \\ 4/5 & 8/5 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -3/5 & 4/5 \\ 4/5 & 3/5 \end{bmatrix}$$

Geometrically, we expect that if we apply a reflection twice, we get back to where we started. Let's check this with the reflection matrix:

$$\begin{bmatrix} -3/5 & 4/5 \\ 4/5 & 3/5 \end{bmatrix}^2 = \begin{bmatrix} -3/5 & 4/5 \\ 4/5 & 3/5 \end{bmatrix} \begin{bmatrix} -3/5 & 4/5 \\ 4/5 & 3/5 \end{bmatrix} = \begin{bmatrix} 9/25 + 16/25 & -12/25 + 12/25 \\ -12/25 + 12/25 & 16/25 + 9/25 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Our next topic will be to study in more generality when we can “undo” a linear transformation like this. Before this, we will summarize what we know about rotations and reflections.

1.2.3 Summary of Rotation and Reflection Matrices

There is a way of writing rotation and reflection matrices so that they have a similar form. Letting $a = \cos(\theta)$, and $b = \sin(\theta)$, a rotation matrix R_θ has the form

$$\begin{bmatrix} a & -b \\ b & a \end{bmatrix}$$

Here a and b satisfy $a^2 + b^2 = 1$. Conversely, any matrix of this form is a rotation. For reflection matrices, if we let

$$a = \frac{w_1^2 - w_2^2}{w_1^2 + w_2^2}$$

$$b = \frac{2w_1 w_2}{w_1^2 + w_2^2}$$

then our formula for a reflection matrix comes out to

$$\begin{bmatrix} a & b \\ b & -a \end{bmatrix}$$

These also satisfy $a^2 + b^2 = 1$. Conversely, any such matrix is a reflection.

2 Inverses Matrices

We can solve linear equations in a single variable

$$ax = b$$

by multiplying both sides by a^{-1} , when $a \neq 0$:

$$x = a^{-1}b$$

We would like to solve linear systems

$$Ax = b$$

in the same way. We need to understand what A^{-1} is, when it exists, and how to compute it.

2.1 Inverses of functions in generality, injectivity and surjectivity

We will approach inverses of matrices from the linear transformation point of view, so we start by considering inverses of functions.

2.1.1 Functions and Their Inverses

In general, we write $f : X \rightarrow Y$ if f takes things in X as inputs and gives things in Y as output, e.g

$$f : \mathbb{R} \rightarrow \mathbb{R}$$

$$f(x) = x^3$$

$$g : \mathbb{N} \rightarrow \mathbb{R}^2$$

$$g(n) = \begin{cases} \begin{bmatrix} 1 \\ 0 \end{bmatrix} & n \text{ is even} \\ \begin{bmatrix} 0 \\ 1 \end{bmatrix} & n \text{ is odd} \end{cases}$$

$$h : \{\text{students in the room}\} \rightarrow \{\text{seats in the room}\}$$

$$h(s) = \text{the seat taken by } s$$

If $f : X \rightarrow Y$ is a function, and $g : Y \rightarrow X$ is a function the other way, we say that g is an inverse of f if for any x , $g(f(x)) = x$, and for any y , $f(g(y)) = y$. We say f is *invertible* if it has an inverse.

Example 5. Our function f above has an inverse $g(x) = x^{1/3}$, the cube root function.

We can characterize invertible functions using the following properties:

Definition. Let $f : X \rightarrow Y$ be a function.

1. f is *surjective* if for any $y \in Y$, there is an $x \in X$ with $f(x) = y$. This says that any equation $f(x) = b$ has at least one solution.
2. f is *injective* if for $x_1, x_2 \in X$, if $x_1 \neq x_2$, then $f(x_1) \neq f(x_2)$. It is equivalent to say that if $f(x_1) = f(x_2)$, then $x_1 = x_2$, this is often easier to check directly. This says that if the equation $f(x) = b$ has a solution, it is unique.

Example 6. Our function f above is both injective and surjective, for any $b \in \mathbb{R}$, there is a unique real number a such that $a^3 = b$. The function g above is neither injective nor surjective. There is no n such that

$$g(n) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

for example, and $f(1) = f(3)$. The function h is injective but (probably) not surjective.

Proposition. A function f is invertible if and only if it is injective and surjective. Its inverse f^{-1} then can be described as

$$f^{-1}(y) = \text{the unique solution } x \text{ to the equation } f(x) = y$$

Question. Does $f(x) = x^2$, $f : \mathbb{R} \rightarrow \mathbb{R}$ have an inverse?

No, it is not injective, $f(-1) = f(1)$. It is also not surjective, there is no real number a with $a^2 = -1$.

Question. Does a rotation of the plane have an inverse?

Yes, a rotation through θ is inverted by a rotation through $-\theta$.

2.2 Inverses of linear transformations are linear

Suppose that $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a linear transformation which is invertible, with inverse T^{-1} . (It will turn out that we cannot have an invertible linear transformation $\mathbb{R}^n \rightarrow \mathbb{R}^m$ for $m \neq n$, for now this can be taken on faith.) This means that for any vector w , the linear system $T(x) = w$ has a unique solution. Then T^{-1} sends a vector w this unique solution.

Claim. T^{-1} is itself linear.

Proof. We wish to show that $T^{-1}(w_1 + w_2) = T^{-1}(w_1) + T^{-1}(w_2)$. Let $v_1 = T^{-1}(w_1)$ and $v_2 = T^{-1}(w_2)$. Then $T(v_1) = w_1$ and $T(v_2) = w_2$.

$$T(v_1 + v_2) = T(v_1) + T(v_2) = w_1 + w_2$$

Thus $v_1 + v_2$ is a solution to $T(x) = w_1 + w_2$, and so must be equal to $T^{-1}(w_1 + w_2)$, because this is the unique solution. Similarly, suppose $v = T^{-1}(w)$, so $T(v) = w$. Then for any k , $T(kv) = kw$, so kv is a solution to $T(x) = kw$, and therefore $kv = T^{-1}(kw)$. $\square$

This tells us that T^{-1} is represented by a matrix. If $T(x) = Ax$, we write A^{-1} for the matrix representing T^{-1} . Finding A^{-1} gives us a way to solve *all* linear systems with coefficient matrix A in a uniform way. The linear system $Ax = b$ has unique solution $x = A^{-1}b$ for any b .

Example 7. Solve

$$R_{\pi/3}(x) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

where $R_{\pi/3}$ is the rotation through $\pi/3$. The inverse of $R_{\pi/3}$ is $R_{-\pi/3}$, which has matrix

$$\begin{bmatrix} \cos(-\pi/3) & -\sin(-\pi/3) \\ \sin(-\pi/3) & \cos(-\pi/3) \end{bmatrix} = \begin{bmatrix} 1/2 & \sqrt{3}/2 \\ -\sqrt{3}/2 & 1/2 \end{bmatrix}$$

and so our solution is

$$\begin{bmatrix} 1/2 & \sqrt{3}/2 \\ -\sqrt{3}/2 & 1/2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1+\sqrt{3}}{2} \\ \frac{1-\sqrt{3}}{2} \end{bmatrix}$$

2.3 Algebraic Properties of Inverse Matrices

Another way to characterize the inverse matrix A^{-1} is with the following property:

$$AA^{-1} = A^{-1}A = I_n$$

This corresponds to the equations $T(T^{-1}(y)) = y$ and $T^{-1}(T(x)) = x$. Using this, we can find a formula for the inverse of a matrix product. Consider two invertible matrices A and B . I claim that $(AB)^{-1} = B^{-1}A^{-1}$. To see this, we compute

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AI_nA^{-1} = AA^{-1} = I_n$$

$$(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}I_nB = B^{-1}B = I_n$$

2.4 When is a matrix invertible?

From the above, an $n \times n$ matrix is invertible if and only if for any b , the equation $Ax = b$ has a unique solution. Thinking back to our method of solving linear systems, this will be true just in case every column in the reduced row echelon form of A is a pivot column. In other words:

Proposition. *An $n \times n$ matrix is invertible if and only if its reduced row echelon form is the identity matrix I_n .*

This gives us a practical way to check if a matrix is invertible. (On the homework, you will see another way to check invertibility for 2×2 matrices. This will be generalized to $n \times n$ matrices for all n later in the course.) We can actually use row operations to compute the inverse when it exists.

2.5 Computing Inverse Matrices

Let A be a matrix corresponding to an invertible transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$. To find A^{-1} , we need to find the matrix representing T^{-1} . The columns of this matrix are the vectors $T^{-1}(e_i)$. We can describe $T^{-1}(e_i)$ as the unique solution to $Ax = e_i$, so it seems that we need to solve n linear systems, for $e_1, \dots, e_n$. We can actually solve all of these simultaneously, and compute A^{-1} in one fell swoop.

The process is as follows: Given A an $n \times n$ matrix, construct an augmented matrix

$$[A \mid I_n]$$

Here, note that there is (probably) more than one column to the right of the line; we have one for each e_i . Then, apply row operations to bring A to reduced row echelon form. If the reduced row echelon form is not I_n , then A is not invertible. If it is, we now have

$$[I_n \mid B]$$

with some $n \times n$ matrix B to the right of the line. This B is exactly A^{-1} . Indeed, the i th column of B is the result of solving the equation $Ax = e_i$.

Example 8. Consider the matrix

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

We compute the inverse as follows:

$$\left[\begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{array} \right] \rightsquigarrow \left[\begin{array}{cc|cc} 1 & 2 & 0 & 1 \\ 2 & 1 & 1 & 0 \end{array} \right] \rightsquigarrow \left[\begin{array}{cc|cc} 1 & 2 & 0 & 1 \\ 0 & -3 & 1 & -2 \end{array} \right] \rightsquigarrow \left[\begin{array}{cc|cc} 1 & 2 & 0 & 1 \\ 0 & 1 & -1/3 & 2/3 \end{array} \right] \rightsquigarrow \left[\begin{array}{cc|cc} 1 & 0 & 2/3 & -1/3 \\ 0 & 1 & -1/3 & 2/3 \end{array} \right]$$

Therefore

$$A^{-1} = \begin{bmatrix} 2/3 & -1/3 \\ -1/3 & 2/3 \end{bmatrix}$$

Let's check this directly:

$$\begin{bmatrix} 2/3 & -1/3 \\ -1/3 & 2/3 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 2/3 & -1/3 \\ -1/3 & 2/3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Example 9. Consider the 3×3 matrix

$$A = \begin{bmatrix} 1 & -1 & 1 \\ 2 & 3 & 4 \\ 3 & 2 & 5 \end{bmatrix}$$

Let's try to compute the inverse:

$$\left[\begin{array}{ccc|ccc} 1 & -1 & 1 & 1 & 0 & 0 \\ 2 & 3 & 4 & 0 & 1 & 0 \\ 3 & 2 & 5 & 0 & 0 & 1 \end{array} \right] \rightsquigarrow \left[\begin{array}{ccc|ccc} 1 & -1 & 1 & 1 & 0 & 0 \\ 0 & 5 & 2 & -2 & 1 & 0 \\ 0 & 5 & 2 & -3 & 0 & 1 \end{array} \right] \rightsquigarrow \left[\begin{array}{ccc|ccc} 1 & -1 & 1 & 1 & 0 & 0 \\ 0 & 5 & 2 & -2 & 1 & 0 \\ 0 & 0 & 0 & -1 & -1 & 1 \end{array} \right]$$

We can see at this point that we will not have enough pivot columns. The conclusion we draw is that the matrix is not invertible.

3 Subspaces, Linear Dependence, and Span

3.1 Spans and Images of Linear Transformations

Given vectors $v_1, \dots, v_r \in \mathbb{R}^n$, their *span* is the set of all linear combinations of them.

$$\text{span}\{v_1, \dots, v_r\} = \{x_1v_1 + \dots + x_nv_n \mid x_1, \dots, x_n \in \mathbb{R}\}$$

If we package the vectors into a matrix

$$A = \begin{bmatrix} \vdots & & \vdots \\ v_1 & \dots & v_r \\ \vdots & & \vdots \end{bmatrix}$$

then this is exactly the set of vectors which can be written Aw for some $w \in \mathbb{R}^r$. This is to say, if $T : \mathbb{R}^r \rightarrow \mathbb{R}^n$ is the linear transformation corresponding to A , the span is the *image* of this function $\text{im}(T)$: the set of all outputs to the function $\{T(w) \mid w \in \mathbb{R}^r\}$.

Example 10. Does the span of

$$v = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \quad w = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

contain the vector

$$u = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} ?$$

We want to know if we can write u as $xv + yw$. This comes down to solving the system of linear equations with augmented matrix

$$\left[\begin{array}{cc|c} 1 & -1 & 2 \\ 2 & 0 & 1 \\ 0 & 1 & 3 \end{array} \right]$$

Applying row operations, we get:

$$\left[\begin{array}{cc|c} 1 & -1 & 2 \\ 2 & 0 & 1 \\ 0 & 1 & 3 \end{array} \right] \rightsquigarrow \left[\begin{array}{cc|c} 1 & 0 & 5 \\ 2 & 0 & 1 \\ 0 & 1 & 3 \end{array} \right] \rightsquigarrow \left[\begin{array}{cc|c} 1 & 0 & 5 \\ 0 & 0 & -9 \\ 0 & 1 & 3 \end{array} \right]$$

The second row now corresponds to $0 = -9$, so the system is inconsistent. We conclude that u is not in $\text{span}\{v, w\}$. We can write $u \notin \text{span}\{v, w\}$.

This suggests another way of looking at the span, or image of a linear transformation. The image of a matrix A , the span of its columns, is the set of vectors w such that the linear system $Ax = w$ is consistent. We can also say that an $m \times n$ matrix A is surjective as a linear transformation if and only if $\text{im}(A) = \mathbb{R}^m$.

3.2 Linear Subspaces in General

The span of a set of vectors, equivalently the image of a linear transformation, has certain closure properties. If v, w are in the image of T , we can write $T(u_1) = v$, $T(u_2) = w$. Then $T(u_1 + u_2) = v + w$, so $v + w$ is also in the image. Similarly, for any $k \in \mathbb{R}$, $T(ku_1) = kv$, so any kv is in the image. In terms of linear combinations, we have

$$(x_1v_1 + \cdots + x_nv_n) + (y_1v_1 + \cdots + y_nv_n) = (x_1 + y_1)v_1 + \cdots + (x_n + y_n)v_n$$

$$k(x_1v_1 + \cdots + x_nv_n) = (kx_1)v_1 + \cdots + (kx_n)v_n$$

Definition. A *linear subspace* of $\mathbb{R}^n$ is a set V of vectors in $\mathbb{R}^n$ with the following closure properties:

1. V contains the 0 vector.
2. If $v, w \in V$, then $v + w \in V$.
3. If $v \in V$ and $k \in \mathbb{R}$, then $kv \in V$.

In $\mathbb{R}$, the only subspaces are $\{0\}$ and $\mathbb{R}$. In $\mathbb{R}^2$, we will see that subspaces are $\{0\}$, $\mathbb{R}^2$ itself, and lines through the origin. In $\mathbb{R}^3$, we have $\{0\}$, $\mathbb{R}^3$, and lines and planes through the origin.

Given $v_1, \dots, v_r \in V$, these vectors are said to *span* V if $\text{span}\{v_1, \dots, v_r\} = V$. This is our first way of describing subspaces: giving a spanning set.

3.3 The Kernel of a Matrix

Consider a linear system of the form $Ax = 0$, for A a $m \times n$ matrix. Such a system will be called *homogeneous*. The set of solutions to this equation is called the *kernel* of A , $\ker(A)$.

Proposition. For any A , $\ker(A)$ is a linear subspace.

Proof. If $Av = 0$ and $Aw = 0$, then $A(v + w) = Av + Aw = 0$, and $A(kv) = kAv = 0$, by linearity. $\square$

This subspace tells us about the non-uniqueness of linear systems $Ax = b$. If w is a solution, then so is $w + v$ for any $v \in \ker(A)$. Moreover, if w_1, w_2 are solutions, then $w_1 - w_2 \in \ker(A)$. Therefore, given a particular solution w , the set of all solutions is exactly

$$\{w + v \mid v \in \ker(A)\}$$

We can conclude from this that A is injective as a linear transformation if and only if $\ker(A) = \{0\}$.

The rows of A can be viewed as equations that must be satisfied by any element of $V = \ker(A)$. This is the second way we have of describing subspaces, by defining equations.

3.4 Translating between different representations of subspaces

Each of these representations is useful for different purposes. If we want to generate lots of elements of space, then we can do this with a spanning set. If we want to decide whether a given element is in the subspace, we can do this by checking defining equations. It is of interest to be able to translate between these representations. We have in some sense already been doing this when we have solved linear systems; I will illustrate this with some examples.

Example 11. Suppose we are given the matrix

$$A = \begin{bmatrix} 1 & -1 \\ 3 & 2 \\ 0 & 4 \end{bmatrix}$$

We would like to find defining equations for the subspace $\text{im}(A)$. In other words, we would like to find equations telling us when a vector

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

is in $\text{im}(A)$. To do this, we consider the augmented matrix

$$\left[\begin{array}{cc|c} 1 & -1 & x \\ 3 & 2 & y \\ 0 & 4 & z \end{array} \right]$$

We can apply row operations to bring this to reduced row echelon form, as follows:

$$\rightsquigarrow \left[\begin{array}{cc|c} 1 & -1 & x \\ 0 & 5 & y-3x \\ 0 & 4 & z \end{array} \right] \rightsquigarrow \left[\begin{array}{cc|c} 1 & -1 & x \\ 0 & 5 & y-3x \\ 0 & 1 & z/4 \end{array} \right] \rightsquigarrow \left[\begin{array}{cc|c} 1 & 0 & x+z/4 \\ 0 & 0 & y-3x-5z/4 \\ 0 & 1 & z/4 \end{array} \right] \rightsquigarrow \left[\begin{array}{cc|c} 1 & 0 & x+z/4 \\ 0 & 1 & z/4 \\ 0 & 0 & y-3x-5z/4 \end{array} \right]$$

The conclusion we draw from this is that the system is consistent, that is, our vector is in $\text{im}(A)$, if and only if $y-3x-5z/4=0$. Translating back into matrix form, we have

$$\text{im}(A) = \ker([-3 \quad 1 \quad -5/4])$$

Example 12. Now consider the matrix

$$A = \begin{bmatrix} -2 & 3 & 1 & 4 \\ 2 & 1 & 0 & -1 \end{bmatrix}$$

We want to find a spanning set for $\ker(A)$. In other words, we want to describe all solutions to $Ax=0$. To this end, we consider the augmented matrix

$$\left[\begin{array}{cccc|c} -2 & 3 & 1 & 4 & 0 \\ 2 & 1 & 0 & -1 & 0 \end{array} \right]$$

The row operations go as follows:

$$\rightsquigarrow \left[\begin{array}{cccc|c} -2 & 3 & 1 & 4 & 0 \\ 0 & 4 & 1 & 3 & 0 \end{array} \right] \rightsquigarrow \left[\begin{array}{cccc|c} 1 & -3/2 & -1/2 & -2 & 0 \\ 0 & 1 & 1/4 & 3/4 & 0 \end{array} \right] \rightsquigarrow \left[\begin{array}{cccc|c} 1 & 0 & -1/8 & -7/8 & 0 \\ 0 & 1 & 1/4 & 3/4 & 0 \end{array} \right]$$

If we interpret this as a system of equations in variables w, x, y, z , we have

$$\begin{cases} w - y/8 - 7z/8 = 0 \\ x + y/4 + 3z/4 = 0 \end{cases}$$

The non-pivot variables y, z can be set arbitrarily, determining the pivot variables:

$$\begin{cases} w = y/8 + 7z/8 \\ x = -y/4 - 3z/4 \end{cases}$$

Thus a general solution has the form

$$\begin{bmatrix} y/8 + 7z/8 \\ -y/4 - 3z/4 \\ y \\ z \end{bmatrix} = y \begin{bmatrix} 1/8 \\ -1/4 \\ 1 \\ 0 \end{bmatrix} + z \begin{bmatrix} 7/8 \\ -3/4 \\ 0 \\ 1 \end{bmatrix}$$

We can conclude that

$$\ker(A) = \text{im} \begin{bmatrix} 1/8 & 7/8 \\ -1/4 & -3/4 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$$

3.5 Linear Dependence and Bases

When we describe a linear subspace by giving a spanning set, some of the vectors may be redundant. For example, if we consider

$$\text{span} \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 4 \end{bmatrix} \right\}$$

we have

$$x \begin{bmatrix} 1 \\ 2 \end{bmatrix} + y \begin{bmatrix} 2 \\ 4 \end{bmatrix} = \begin{bmatrix} x + 2y \\ 2x + 4y \end{bmatrix} = (x + 2y) \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

Therefore, this subspace can also be represented as just

$$\text{span} \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\}$$

Geometrically, both vectors lie on the same line [draw picture]. In this case, this happens because

$$\begin{bmatrix} 2 \\ 4 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} \in \text{span} \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\}$$

Another way of saying this is that these vectors satisfy the relation

$$2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} - \begin{bmatrix} 2 \\ 4 \end{bmatrix} = 0$$

Proposition. *Given a collection of vectors $v_1, \dots, v_r \in \mathbb{R}^n$, the following are equivalent.*

1. $\text{span}\{v_1, \dots, v_r\}$ can also be written as $\text{span}\{v_1, \dots, v_{i-1}, v_{i+1}, \dots, v_r\}$, omitting some v_i .
2. Some v_i is in the span of the other vectors.
3. There is a linear relation

$$x_1 v_1 + \dots + x_n v_n = 0$$

where some $x_i \neq 0$.

Proof. To see that (1) and (2) are equivalent, we notice that if v_i can be omitted, then it has to be in the span of the other vectors, because $v_i \in \text{span}\{v_1, \dots, v_r\}$. Conversely, if v_i is in the span of the other vectors, say

$$v_i = \sum_{j \neq i} a_j v_j$$

Then in any linear combination

$$x_1 v_1 + \dots + x_i v_i + \dots + x_n v_n$$

we can replace v_i ,

$$x_1 v_1 + \dots + x_i \left(\sum_{j \neq i} a_j v_j \right) + \dots + x_n v_n$$

which comes out to a linear combination of the remaining vectors. Thus v_i is redundant.

To see that (2) and (3) are equivalent, if v_i is a linear combination of the other vectors, we can rearrange this to get an equation like in (3). Conversely, if (3) holds with $x_i \neq 0$, then we can solve for v_i , expressing it as a linear combination of the other vectors. $\square$

If any/all of these conditions hold, we say that the vectors $v_1, \dots, v_r$ are *linearly dependent*. If not, we say that they are *linearly independent*.

An important feature of linearly independent collections of vectors is that any element of their span has a *unique* representation as a linear combination. Indeed, if

$$x_1 v_1 + \dots + x_r v_r = y_1 v_1 + \dots + y_r v_r$$

then

$$(x_1 - y_1)v_1 + \dots + (x_r - y_r)v_r = 0$$

and so each $x_i - y_i = 0$, that is, $x_i = y_i$.

Conclusion. For vectors $v_1, \dots, v_r$, if we put them into a matrix

$$A = \begin{bmatrix} \vdots & & \vdots \\ v_1 & \dots & v_r \\ \vdots & & \vdots \end{bmatrix}$$

Then $v_1, \dots, v_r$ are linearly independent if and only if A is injective, if and only if $\ker(A) = \{0\}$.

To see this, notice that linear relations $x_1 v_1 + \dots + x_r v_r$ correspond to vectors

$$\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \in \ker(A)$$

The kernel being 0 means that there are no nontrivial relations. The matrix being injective is exactly the uniqueness remarked above.

If $V = \text{span}\{v_1, \dots, v_r\}$ is a subspace and $v_1, \dots, v_r$ are linearly independent, then we say that $v_1, \dots, v_r$ form a *basis* of V . In this case, any element of V can be expressed in exactly one way as a linear combination of $v_1, \dots, v_r$. In particular, if $\text{span}\{v_1, \dots, v_r\} = \mathbb{R}^n$ and $v_1, \dots, v_r$ are linearly independent, we say that these vectors form a basis for $\mathbb{R}^n$. (We will see that in this case it must be that $r = n$). We have already been working with a basis for $\mathbb{R}^n$, consisting of the vectors $e_1, \dots, e_n$.

Many other bases exist. In fact

Proposition. Vectors $v_1, \dots, v_n \in \mathbb{R}^n$ form a basis if and only if the $n \times n$ matrix

$$\begin{bmatrix} \vdots & & \vdots \\ v_1 & \dots & v_n \\ \vdots & & \vdots \end{bmatrix}$$

is invertible.

Proof. This is a translation of the fact that a matrix is invertible if and only if it is injective, meaning that the columns are linearly independent, and surjective, meaning that the span of the columns is all of $\mathbb{R}^n$. $\square$

Therefore, the vectors

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

form a basis for $\mathbb{R}^2$, for example. You saw another basis for $\mathbb{R}^2$ on the first homework. If we are given a spanning set for a subspace

$$V = \text{span}\{v_1, \dots, v_r\}$$

then we can convert it into a basis using the following procedure. If it is not already a basis, then one vector v_i must be redundant, it can be written as a linear combination of the others. Remove this one. Proceed until no vectors are redundant.

This is simple to describe, but to apply it we need to be able to decide if a given vector is redundant. This requires us to solve a linear system to determine if the vector is in the span of the others, so if we need to do this many times, this can become computationally intense. There is an easier method, which allows us to find a basis by applying Gaussian elimination a single time.

Given vectors $v_1, \dots, v_r$, we put them into a matrix

$$A = \begin{bmatrix} \vdots & & \vdots \\ v_1 & \dots & v_r \\ \vdots & & \vdots \end{bmatrix}$$

We then bring this to reduced row echelon form:

$$B = \begin{bmatrix} \vdots & & \vdots \\ w_1 & \dots & w_n \\ \vdots & & \vdots \end{bmatrix}$$

The first thing to observe is that the pivot columns of B form a basis for $\text{span}\{w_1, \dots, w_n\}$. Indeed, this span is exactly the collection of vectors whose only nonzero entries are in the position of nonzero rows of B . We will argue that the corresponding columns of A form a basis of its image. The point is that the kernel of A and the kernel of B are the same. The kernels are the solution sets of $Ax = 0$, $Bx = 0$ respectively, so this is to say that row operations do not change the solution set of a linear system.

Because the kernel records all linear relations between the columns, this implies that the columns of A which become pivot columns are linearly independent, and that any additional column can be written as a linear combination of them.

Example 13. Find a basis for the span of the following vectors in $\mathbb{R}^3$:

$$v_1 = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, v_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, v_3 = \begin{bmatrix} 3 \\ -1 \\ 2 \end{bmatrix}, v_4 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

We package the vectors into a matrix and bring to reduced row echelon form:

$$\begin{bmatrix} 1 & -1 & 3 & 1 \\ 1 & 1 & -1 & 2 \\ 2 & 0 & 2 & 3 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & -1 & 3 & 4 \\ 0 & 2 & -4 & 1 \\ 0 & 2 & -4 & 1 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & -1 & 3 & 4 \\ 0 & 2 & -4 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & -1 & 3 & 4 \\ 0 & 1 & -2 & 1/2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & 0 & 1 & 9/2 \\ 0 & 1 & -2 & 1/2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Here the first and second columns are pivots. This tells us that v_1, v_2 form a basis for the subspace.

3.6 Dimension and Rank

From what we know so far, it is conceivable that we could find one basis for a subspace of $\mathbb{R}^n$ consisting of 2 vectors, and another consisting of 3 vectors, say. This would in effect give us two coordinate systems on the subspace, one where we use 2 coordinates, and other where we use 3. In this hypothetical world, it would not make sense to say that we have 2 degrees of freedom when choosing an element of the subspace, because we could just as well say that we have 3.

Thankfully, this does not happen, and the notion of “number of degrees of freedom” not only makes sense, it gives us a useful way of describing the size of vector subspaces.

Theorem 1. *For a subspace $V \subset \mathbb{R}^n$, any two bases of V have the same size.*

Proof Sketch. This comes down to the following statement. If $v_1, \dots, v_r$ are linearly independent elements of V , and $w_1, \dots, w_s$ span V , then $r \leq s$. If both the v_i and the w_j are bases, then we can apply this in both directions to get $r = s$. To prove the claim, we can write

$$v_i = \sum_j a_{ij} w_j$$

Let A be the matrix with entries a_{ij} . If $r > s$, then this matrix has more columns than rows, so $Ax = 0$ must have a nonzero solution; it represents a system with more variables than equations. This means that there are $x_1, \dots, x_r$, not all 0, such that

$$\sum_i a_{ij} x_i = 0$$

for each fixed j . But this means that

$$\sum_i x_i v_i = \sum_i \sum_j x_i a_{ij} w_j = \sum_j \left[\sum_i x_i a_{ij} \right] w_j = 0$$

This contradicts the linear independence of the v_i . □

Knowing this, we define the *dimension* of a subspace V to be the size of any basis. In particular, $\mathbb{R}^n$ has dimension n , as we expect. This gives us a way to see that an invertible matrix must be square. If a matrix is invertible and has n rows, then its columns form a basis for $\mathbb{R}^n$, so there must be n of them.

If A is a matrix, then $\text{im}(A)$ is a subspace. We define the rank of A to be $\text{rk}(A) = \dim(\text{im}(A))$. Above, we introduced a method of computing a basis for $\text{im}(A)$, taking columns corresponding to pivot columns in the reduced row echelon form.

Conclusion. $\text{rk}(A)$ is the number of columns in the reduced row echelon form of A .

This gives us an effective way to compute the dimension of subspaces in terms of a spanning set.

Example 14. Find the dimension of

$$\text{span} \left\{ \begin{bmatrix} -2 \\ 2 \\ 6 \end{bmatrix}, \begin{bmatrix} 3 \\ -3 \\ -9 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ -3 \end{bmatrix} \right\}$$

The row operations go as follows:

$$\begin{bmatrix} -2 & 3 & 1 \\ 2 & 3 & 1 \\ 6 & -9 & -3 \end{bmatrix} \rightsquigarrow \begin{bmatrix} -2 & 3 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & -3/2 & -1/2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

We have a single pivot column, so the dimension is 1.

Previously, given a matrix, we saw how to write down a spanning set for its kernel. To do this, we solve the linear system $Ax = 0$ using Gaussian elimination. In reduced row echelon form, each non-pivot column gives us a free variable, and these let us write any element of the kernel as a linear combination of certain vectors. It turns out that these vectors are always linearly independent, and so form a basis for the kernel.

The reason for this is best illustrated with an example. Let's look at a matrix which is already in reduced row echelon form.

$$A = \begin{bmatrix} 1 & -3 & 0 & 2 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

If we translate $Av = 0$ into a linear system in variables w, x, y, z , w and y are pivot variables, and x and z are non pivots. We can solve for w and y :

$$w = 3x - 2z$$

$$y = 2z$$

and hence the general solution to $Av = 0$ is

$$\begin{bmatrix} 3x - 2z \\ x \\ 2z \\ z \end{bmatrix} = x \begin{bmatrix} 3 \\ 1 \\ 0 \\ 0 \end{bmatrix} + z \begin{bmatrix} -2 \\ 0 \\ 2 \\ 1 \end{bmatrix}$$

In this case we can see that the two vectors are linearly independent by looking at the second and fourth entries. If the linear combination is to be 0, then we must have $x = 0$, because the second entry is just x , and $z = 0$, because the fourth entry is just z . Because of the way we assign the free variables to non-pivot columns, this always happens, for each variable, there is an entry in the vector where it appears alone.

We can conclude from this that $\dim(\ker(A))$ is the number of non-pivot columns in the reduced row echelon form of A . Because $\dim(\text{im}(A))$ is the number of pivot columns, we get the following.

Theorem 2 (Rank Nullity Theorem). *If A is an $m \times n$ matrix, representing a transformation $\mathbb{R}^n \rightarrow \mathbb{R}^m$, then*

$$\dim(\text{im}(A)) + \dim(\ker(A)) = n$$

The reason for the name is that $\dim(\text{im}(A))$ is the rank of A , and $\dim(\ker(A))$ is sometimes called the nullity.